

quate despite optimal dosage, tracheostomy remains the treatment of choice. In these circumstances and in cases of severe life-threatening obstructive sleep apnea, the benefits of tracheostomy clearly outweigh the risks. However, when the disorder appears benign or when tracheostomy is refused, protriptyline can be appropriate and effective treatment.

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CIGARETTE SMOKING AS A RISK FACTOR FOR EPIDEMIC A(H₁N₁) INFLUENZA IN YOUNG MEN

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Abstract We studied an outbreak of A(H₁N₁) influenza in an Israeli military unit of 336 healthy young men to determine the relation of cigarette smoking to the incidence of clinically apparent influenza and to the influenza-antibody response. Of 168 smokers, 68.5 per cent had influenza, as compared with 47.2 per cent of nonsmokers ($P < 0.0001$). Influenza was also more severe in the smokers; 50.6 per cent of the smokers lost work days or required bed rest, or both, as compared with 30.1 per cent of the nonsmokers. The proportion of all influenza in smokers that was attributable to smoking was 31.2 per cent (95 per

cent confidence intervals, 16.5 to 43.1 per cent). For severe influenza, the attributable risk in the smokers was 40.6 per cent (95 per cent confidence intervals, 21.6 to 54.8 per cent). A quarter of all severe morbidity from influenza in the overall study population was attributable to smoking. Antibody levels to A/USSR/90/77(H₁N₁) antigen were higher in smokers but not markedly so.

We conclude that smoking is a major determinant of morbidity in epidemic influenza and may contribute substantially to incapacitation in outbreaks in populations that smoke heavily. (*N Engl J Med*. 1982; 307:1042-6.)

THE association between smoking and epidemic influenza has been the subject of only limited investigation, although there is some evidence that smokers tend to have excess morbidity from influenza.¹⁻³ Smokers among female recruits in the Israel Defence Forces had a 44 per cent increase in morbidity during an epidemic influenza-like illness suspected to be caused by the A/USSR/77(H₁N₁) subtype.⁴ Since both smoking and epidemic influenza are widespread, the impact of a causal association between the two would be of immense importance in terms of health and economic considerations. The objective of this study was to examine the relation between cigarette smoking and clinical and serologic manifestations of epidemic influenza in an otherwise healthy population

of young men, and to estimate the proportion of morbidity from influenza — both in the smokers and in the entire study population — that was attributable to smoking.

METHODS

The Outbreak

The study population and the clinical and serologic characteristics of the outbreak have been described elsewhere.⁵ The population at risk consisted of 336 healthy male recruits aged 18.5 ± 0.7 years (mean $\pm$ S.D.). Military service is obligatory in Israel. The ethnic and educational characteristics of the study population resemble those of the total male age cohort. A clearly delineated outbreak of influenza occurred in the winter of 1978 over a 17-day period. The symptoms were generally mild; fever exceeded 38°C in half the cases.

Identification of the Virus

Nasopharyngeal swabs taken from 22 acutely ill patients were placed in tubes with M-199 medium containing antibiotics. These were inoculated into monkey kidney-cell cultures and into 10-day-old fertilized eggs. An influenza A virus, similar to A/USSR/90/77(H₁N₁) was isolated in 5 of the 22 patients. We found this virus considerably more difficult to isolate than previous subtypes. After treatment to remove nonspecific inhibitors, paired serum samples from these patients, obtained during the acute and five-week conva-

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